

Medical Equipment Utilization & Maintenance Risk Monitoring System for Kenyan County Hospitals

Predictive maintenance and usage accountability for critical medical equipment

1. Team Members and Roles

Name	Role	Responsibility
Brenda Chepkirui	Solutions Engineer / Data Analyst	Problem definition, data strategy, analytics workflow, business translation
Valentine Nyankwoga	Data Engineer	Data pipeline, ingestion, feature engineering, database design
Caleb Munyekenye	ML Engineer	Risk-prediction model development and evaluation
Paul Omondi	Dashboard / Frontend Engineer	Facility risk dashboard and alerting interface
Evans Macharia	DevOps Engineer	CI/CD, environment management, deployment, integration, versioning

2. Problem Statement

In Kenya's public county hospital system, the problem of undetected equipment breakdown risk, prolonged idle time, and unreliable usage reporting for critical medical equipment — dialysis machines, ICU equipment, imaging and radiotherapy machines — leads to interrupted life-sustaining care, wasted public investment, and weakened financial accountability over a multi-billion-shilling national equipment program.

This is a documented, current problem, not a hypothetical one. A 2026 national audit of Kenya's Managed Equipment Services (MES) dialysis program, which examined hospital records and machine usage data from 2015 to 2021, found that in almost eight out of ten hospitals, the machines' own usage counters showed far more work performed than what was recorded in the

hospitals' paper registers — in some cases by more than 30 percent. Separately, machines across 125 hospitals under the same Sh38–63 billion scheme ground to a halt due to lack of maintenance, prompting government termination of the contracts, while multi-billion-shilling equipment including CT scanners and mammogram machines sit idle in several counties due to staffing or infrastructure gaps. The problem is current: in January 2025 a dialysis machine breakdown at Naivasha sub-county hospital forced patients to seek care elsewhere, and in 2025 Kenyatta National Hospital's main cancer radiotherapy machine failed unexpectedly, disrupting oncology services nationally.

The gap is a monitoring and accountability problem, not an absence of underlying data.

Hospitals already log equipment usage, whether through machine counters or paper registers, and the Ministry of Health's Kenya Health Facility Census already records facility-level equipment inventory and service availability nationally. What is missing is a systematic way to compare recorded usage against expected or counter-verified usage, flag facilities where equipment sits idle or is at elevated breakdown risk, and shift maintenance from reactive (fix after failure) to condition-based (intervene before failure) — exactly the kind of gap the 2026 MES audit exposed after the fact, at national scale.

Currently, equipment monitoring relies on manual paper registers and reactive maintenance triggered only after a breakdown has already disrupted patient care. This approach fails because paper-based reporting is demonstrably unreliable at scale, as the counter-versus-register discrepancy shows, and because reactive maintenance means a machine is only inspected once it has already stopped working — by which point patients have already been turned away or delayed.

Proposed approach: use data engineering and machine learning to combine facility-level equipment inventory data, usage/maintenance logs, and documented incident patterns into a risk-monitoring system that flags equipment at elevated breakdown risk and highlights usage-reporting discrepancies, aiming to shift facilities from reactive to condition-based maintenance and improve accountability over public equipment investment.

3. Domain & Context

Industry: Healthcare — public hospital medical equipment asset management

Specific Operational Context: Kenya's county hospital network operating under the national Managed Equipment Services (MES) program and its successor, the National Equipment Support to Counties (NESC) scheme — covering dialysis, ICU, radiology, and theatre equipment leased or supplied to public hospitals across all 47 counties. This is critical care infrastructure: dialysis and ICU equipment in particular are life-sustaining, and an unplanned outage directly and immediately affects patient survival, not just service quality.

The operational challenge sits at the intersection of asset integrity monitoring (predicting breakdown before it happens) and financial/usage accountability (reconciling actual usage against what is reported) — both surfaced by the same 2026 national audit, and both

addressable from the same underlying usage and maintenance data, which is what makes a single MVP viable across both angles.

4. Data Source

Real, facility-level equipment usage and maintenance logs are not bulk-published by Kenya's Ministry of Health or individual county hospitals. The data strategy therefore separates real, publicly available context and method-validation data from a small, clearly-labeled set of incident-grounded scenarios used for evaluation.

Layer 1 — Real data, for context and method validation

- Kenya Health Facility Census (Ministry of Health): a real, national government dataset recording facility-level equipment inventory, service availability, and infrastructure gaps — used to build a realistic facility and equipment master list and to derive features such as equipment type, county, and reported service availability.
- A public equipment-degradation telemetry dataset used purely to validate the breakdown-risk prediction method — rolling usage/condition features leading up to failure — on real sensor degradation data, independent of anything the team constructs. This is a method check, not a healthcare-specific claim.

Layer 2 — Incident-grounded scenario data, for the demo

Simulated facility-level equipment usage and maintenance logs constructed to reproduce two specific, documented patterns from the Kenyan case history:

- A usage-versus-reporting discrepancy pattern, modeled on the 2026 MES audit finding of counter-vs-register mismatches
- A maintenance-neglect-to-breakdown pattern, modeled on the Naivasha and MES 125-hospital equipment-halt cases

Each scenario is generated with a written rationale tracing it to a specific cited case, so the dataset is auditable rather than arbitrary.

What this data strategy explicitly does not claim: the project will not present its risk scores as validated against real Kenyan hospital equipment outcomes at scale, since real facility usage/maintenance logs were not accessible. It will claim the underlying method is validated on real degradation data (Layer 1) and demonstrated against realistic, cited Kenyan scenarios (Layer 2) — a distinction stated explicitly in the final presentation, consistent with the standard applied to every other option considered for this capstone.

5. Success Metrics

Technical metrics (model performance)

Metric	Target	Measured against
Breakdown-risk detection rate on labeled scenario cases	> 90%	Scenario dataset (Layer 2)
Lead time before predicted failure window	Flag risk at least 5–10 usage-days before simulated breakdown	Scenario dataset (Layer 2)
Usage-reporting discrepancy detection rate	> 85% of injected discrepancy cases flagged	Scenario dataset (Layer 2)

Lead time before failure is the headline metric, not accuracy alone — it is the number that maps directly to the business case: under reactive maintenance, a facility only learns a machine has failed once it has already stopped working, so any demonstrated lead time is the core result worth defending.

Business metrics (impact framing)

Area	Expected impact if deployed
Patient safety	Fewer unplanned outages of life-sustaining equipment (dialysis, ICU), reducing forced patient transfers and treatment delays
Financial accountability	Usage-reporting discrepancies flagged systematically, supporting oversight of the KSh38–63 billion national equipment investment
Equity of access	Idle or underused equipment surfaced for reallocation, addressing the documented pattern of multi-billion-shilling equipment sitting unused in some counties
Maintenance efficiency	Shift from reactive (post-failure) to condition-based (pre-failure) maintenance scheduling

As with the technical metrics, these are framed as potential impact given the MVP is built on public census data and a labeled scenario set, not a live hospital deployment — the presentation should state this plainly.

6. Business Value

6.1 The size of the opportunity

This system targets a specific, quantified gap inside a national program: Kenya's Managed Equipment Services (MES) scheme represents a Sh38–63 billion public investment, and a 2026 national audit found usage-reporting discrepancies of up to 30% across nearly eight in ten hospitals examined. That gap — the difference between what the government believes is happening with its equipment investment and what is actually happening — is the addressable opportunity: better visibility directly supports better-targeted maintenance spend, fewer catastrophic failures, and a stronger evidence base the next time the program goes back to the regulator or Treasury for funding.

6.2 Illustrative value, benchmarked against predictive maintenance industry data

No Kenyan hospital-equipment-specific ROI study exists yet, so the figures below are illustrative, drawn from cross-industry predictive maintenance research, and should be treated as a directional case for the value of the approach — not a guaranteed local outcome. They are the numbers the pilot is designed to test and validate.

Value Driver	Cross-Industry Benchmark	Source
Reduction in unplanned downtime from reactive → predictive maintenance	30–50%	McKinsey & Company predictive maintenance research
Reduction in overall maintenance cost	10–40% (conservative estimates cluster at 10–20%)	McKinsey & Company; Deloitte
Cost premium of emergency/reactive repair vs. planned maintenance on the same asset	4–5× more expensive	Deloitte

Applied cautiously to a single documented case — the Sh38–63bn MES program — even a conservative 10–20% improvement in maintenance targeting and a reduction in emergency repair premiums represents a materially significant sum in absolute terms, and is the kind of number a pilot with real usage data can actually produce and defend, rather than assert up front.

7. Government Implementation & Rollout Plan

Government rollout is phased deliberately — credibility with a single facility is the precondition for credibility with a county, and county credibility is the precondition for national engagement. Skipping a phase is the most common way public-sector pilots stall.

Phase	Scope	Key Government Touchpoint	Goal
Phase 1: Single-facility pilot	One hospital, one equipment type (dialysis)	Hospital biomedical engineering / facilities management team	Validate the model against real usage and maintenance logs; produce a defensible before/after case
Phase 2: County rollout	2–4 facilities within one county	County Department of Health, County Biomedical Engineering Unit	Prove the model generalizes across facilities with different equipment ages and conditions
Phase 3: National integration	Alignment with the national equipment register	Ministry of Health — Directorate of Health Infrastructure; NESC program management office	Integrate with national MES/NESC oversight and the Kenya Health Facility Census as a standing accountability tool

7.1 Regulatory & data governance considerations

- Data-sharing MOUs with each pilot facility and county before any usage/maintenance data is accessed
- Compliance with Kenya's Data Protection Act — asset-level data carries far lower sensitivity than patient data, which keeps this pathway comparatively fast
- Alignment with existing Ministry of Health data standards so the system is additive to, not competing with, the Kenya Health Facility Census

8. Go-to-Market Strategy

8.1 Customer segments

Segment	Role	Priority
National government (Ministry of Health / NESC program office)	Primary long-term customer — owns the equipment program and the accountability mandate	Primary
County governments / County Health Departments	Operational customer — manages day-to-day facility equipment	Primary
MES/NESC equipment vendors and maintenance contractors	Potential secondary customer — incentivized to prove uptime and contract performance	Secondary

8.2 Business model options

- Government (B2G) contract or tender: a facility- or county-level subscription funded through the existing MES/NESC budget line
- Outcome-based contract: fees tied to demonstrated downtime reduction or reconciled usage-reporting accuracy — aligns incentives and is easier to justify politically than a flat license fee
- Donor / health-systems-strengthening grant funding: global health infrastructure funders regularly fund exactly this kind of accountability tooling as part of broader health system strengthening programs

8.3 Channels

- Direct engagement with County Biomedical Engineering Units, who feel this problem daily and are natural early champions
- Introductions via the sponsoring program (see Section 9) into government and parastatal networks
- Kenya's digital health and health-innovation ecosystem (health innovation forums, health system strengthening implementers) for visibility and second-order introductions

8.4 Competitive landscape

No Kenya-specific competitor targeting this exact gap — usage-vs-reporting reconciliation plus breakdown-risk prediction for public hospital equipment — was identified during this team's research. Indirect competition comes from generic international biomedical asset-management and CMMS (computerized maintenance management system) vendors, which are typically

priced and designed for large private hospital systems rather than resource-constrained public county hospitals, leaving a real gap for a locally-built, appropriately-scoped alternative.

8.5 Early traction goal

Secure a signed pilot MOU with one hospital's biomedical engineering department within three months of program completion — the single most important go-to-market milestone, since every later phase depends on having one real, referenceable deployment.

9. KPC's Role as Program Sponsor

KPC's sponsorship of this program is understood to be at the cohort level — funding and mentoring startups generally, independent of industry — rather than KPC being a customer or user of MasterMaisha itself. That distinction shapes what is being asked for below: program-level accelerator support, not a pipeline-sector partnership.

Even outside its own sector, KPC brings transferable value as a mature Kenyan state corporation: it has direct experience with government procurement, parastatal relations, and large-scale asset-monitoring systems (its own pipeline SCADA and integrity-monitoring operations) — all directly relevant to MasterMaisha's own government go-to-market path, even though the target customer sits in a different ministry.

10. Our Ask from KPC

Ask	Detail	Why It Matters
Seed grant / program funding	Cover pilot-phase costs: data acquisition, compute, and facility engagement logistics for Phase 1	The single biggest constraint on reaching a real pilot MOU is funded runway to do the work before revenue exists
Named technical mentor(s)	A KPC engineer or data/analytics lead with asset-monitoring or SCADA background, for periodic technical review	Directly transferable expertise in monitoring physical asset health at scale, even across sectors
Formal letter of support / introduction	An introduction to the Ministry of Health's Directorate of Health Infrastructure and/or one pilot county's Department of Health	A credible third-party sponsor materially shortens the trust-building

		time with a first government pilot partner
Demo day / investor showcase slot	Inclusion in the program's standard cohort showcase	Visibility to potential co-funders (health-focused investors, donors) beyond the program itself
Post-program incubation support (3–6 months)	Continued access to mentorship and workspace while the Phase 1 pilot MOU is pursued	The pilot MOU will not close within the six-week capstone window; a short bridge period is what turns the capstone into a real pilot

11. Initial Timeline (Weeks 7–12)

Week	Focus	Deliverable
7	Problem framing & data acquisition — finalize MVP scope to one equipment type (dialysis, given the richest documented case history); acquire Kenya Health Facility Census data and the C-MAPSS degradation dataset; design scenario logic tied to cited cases	Data architecture document + initial dataset extracts
8	Data engineering pipeline — clean and join facility census data, build the usage/maintenance log schema, ingest C-MAPSS data	Working ingestion pipeline producing a unified facility-equipment dataset
9	Feature engineering & scenario dataset build — usage trends, maintenance interval features, reporting-discrepancy features; finalize labeled scenario set	Feature-engineered dataset (Layer 1 + Layer 2) ready for modeling

10	Model development — train and tune breakdown-risk model and discrepancy-flagging logic, evaluate against target metrics	Trained risk model with documented evaluation results
11	Dashboard & alerting — facility/equipment risk view, discrepancy flags, lead-time-to-failure display, wired to model output	Working risk dashboard demoing prioritized facility/equipment risk
12	Final testing & presentation — end-to-end validation, business narrative tied to the 2026 MES audit and documented breakdown cases, rehearsal	Final capstone presentation and demo

This timeline reflects a single, focused MVP scoped to one equipment type (dialysis machines) with one combined risk-and-discrepancy capability — sized realistically for a four-person team across six weeks. Extending to additional equipment types (ICU, imaging, radiotherapy) is explicitly named as future work, not attempted this term.